

NETWORKS AND COMPLEXITY

Solution 21-2

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 21.2: Random derivation [Lvl. 2]

Consider a random walk on a network described by an adjacency matrix $\mathbf{A}$. A walker that in a node i leaves at a random point in time such that the departure rate equals the node degree k_i . When leaving the walker picks a link connecting to i at random and moves instantaneously to the link's other endpoint j . Derive a differential equation that governs the dynamics of x_i , the probability that the walker is in node i .

Solution

Let's consider a small time interval dt since the walker leaves at rate k_i . The walker is currently in node i with probability x_i . If it is in node i , it leaves at a rate k_i . Therefore, the total probability that the walker leaves node i in the time interval dt is

$$R_{\text{leave},i} = x_i k_i dt \quad (1)$$

Next consider the rate at which the walker is arriving from a specific neighbor j of node i . The rate at which the walker leaves j is

$$R_{\text{leave},j} = x_j k_j dt \quad (2)$$

If the walker leaves node j , which has degree k_j , the chance that it chooses the link to i is $1/k_j$. Hence, the probability that we see the walker traveling from j to i in the time interval dt is

$$R_{\text{travel},j \rightarrow i} = \frac{x_j k_j}{k_j} dt = x_j dt \quad (3)$$

Since dt is short, we can simply add up the arrival rates from all neighbors of node i ,

$$R_{\text{arrive},i} = \sum_{[ij]} R_{\text{travel},j \rightarrow i} \quad (4)$$

$$= \sum_j A_{ij} R_{\text{travel},j \rightarrow i} \quad (5)$$

$$= \sum_j A_{ij} x_j dt \quad (6)$$

Note that in the arrival term we did not have to multiply an extra factor $(1 - x_i)$ to exclude that the walker is in i already, like we did in the SIS model where we had to exclude that the node is already infected. The difference arises because many nodes can be infected in the epidemic, but in the random walk there is only one walker. Hence, the probabilities are not independent. If the walker is in node j we can be sure that it is not in node i . So, if we look at the case where the walker is currently in j , the probability that it is not in i is not $1 - x_i$ but it is actually 1.

The change in x_i that occurs over the interval dt is the difference of arrival and departure rates,

$$dx_i = R_{\text{arrive},i} - R_{\text{leave},i} = \left(\sum_j A_{ij} x_j dt \right) - (x_i k_i dt) \quad (7)$$

Dividing by dt yields

$$\frac{dx_i}{dt} = \dot{x}_i = -x_i k_i + \sum_j A_{ij} x_j \quad (8)$$

which is the equation that we used as a starting point in the chapter. We recognize this as the i -th component of the vector equation

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{K})\mathbf{x} = -\mathbf{L}\mathbf{x}. \quad (9)$$